

Master Game Design Document (GDD) & Implementation Specification

Project: WebXR Boxing Coach VR (Meta Quest 3)

1. Executive Summary & Vision

* **Target Hardware:** Meta Quest 3 (Standalone Browser / WebXR).

* **Core Technology Stack:** WebXR Device API, Three.js (or WebXR-compatible 3D engine), Web Audio API, Web Haptics API.

* **Core Vision:** A zero-friction, browser-based VR boxing game that balances true athletic coaching with engaging game feel. Players enter a gritty old-school gym, learn authentic boxing fundamentals from an interactive NPC coach, and get an intense cardiovascular workout.

* **Primary Differentiators:**

1. No app store installation required (WebXR link launch).
2. Realistic focus-mitt mitt work and dynamic head-movement mechanics (6DoF).
3. Integrated fitness tracking without immersion-breaking UI.
4. Dual mode: Full Virtual Reality (VR) and Mixed Reality (MR / Passthrough "Garage Mode").

2. Technical Stack & Performance Benchmarks

* **Target FPS:** Solid 90 FPS (minimum 72 FPS fallback) on Meta Quest 3 standalone browser.

* **Libraries:**

* `Three.js` (Rendering & WebXR session management).

* `cannon-es` or `rapier3d` (Lightweight physics for heavy bag collision and glove impact).

* `Web Audio API` (Spatial audio positioning, dynamic music layering, SFX).

* **Controller Tracking:**

* 6DoF hand tracking via Quest 3 Touch Plus controllers.

- * Haptic feedback via `gamepad.hapticActuators` or `pulse()` WebXR API.

* **Rendering Constraints:**

- * Low-poly baked lighting / lightmaps for the gym environment.

- * Maximum 150k total polygons in scene.

- * Single draw-call optimization using texture atlases where possible.

3. Customer Experience & Environment Architecture

3.1 Environment Design

1. **Old-School Gym Mode (VR):**

- * Gritty warehouse setting: brick walls, concrete floors, floor fan, dim overhead spotlights, boxing ring in background.

- * Spatialized ambient soundscape: floor fan hum, jump-rope rhythm in distance, distant heavy bag hits.

2. **Garage Mode (MR / Passthrough):**

- * Leverages Quest 3 passthrough video feed.

- * Renders only the heavy bag, coach NPC, and gym chalk-board mat anchored in the player's real-world room.

3.2 Coach NPC Persona & Behaviors

- * **Appearance:** Veteran boxing trainer with mitts and hand wraps.

- * **Positioning:** Dynamically circles the heavy bag or player. Maintains gaze tracking (eye contact) using IK (Inverse Kinematics).

- * **Voice & Persona:** Encouraging, firm, focused on guard safety and proper technique.

* **Interactions:**

- * Taps player controllers with virtual mitts if guard drops below chin level.

- * Demonstrates punches with slow-motion visual trail ghosts.

- * Holds focus mitts dynamically during pressure tests.

4. Game Mechanics & Fitness Systems

4.1 Punch Numbering System

- * **1:** Lead Jab
- * **2:** Rear Cross
- * **3:** Lead Hook (Head)
- * **3b:** Lead Hook (Body / Liver Shot)
- * **4:** Rear Hook
- * **5:** Lead Uppercut
- * **6:** Rear Uppercut
- * **1b / 2b:** Straight Body Jab / Body Cross

4.2 Head Movement & Footwork Physics

- * **Slips:** Head moves off center line left/right (detected via headset \$X\$-axis offset).
- * **Rolls/Ducks:** Head lowers below horizontal threshold in a "U" trajectory (detected via headset \$Y\$-axis drop).
- * **Pivots:** Body turns 45 degrees relative to origin (tracked via headset yaw and controller vectors).
- * **Level Changes:** Requires physical squatting; firing body shots from standing height invalidates punch accuracy.

4.3 Invisible Stamina & Exertion Engine

- * **Velocity Tracking:** Real-time calculation of controller vector speed:

$$\$V_{\text{punch}} = \frac{\Delta \text{Position}}{\Delta \text{Time}}\$$$

- * **Exertion Accumulation:** High-velocity punches build exertion. Extended continuous punching causes:

1. Heavy breathing audio playing through Web Audio API.

2. Haptic decay (punches feel soft and dull in controller rumble).

3. Audio dampening on bag impact sound (from sharp *crack* to soft *thud*).

* **Active Recovery:** Restores stamina when the player lowers punch volume, maintains guard, and executes defense/slips for 5–10 seconds.

4.4 The Chalkboard UI (Post-Workout Analytics)

An in-world dusty chalkboard on the gym wall displays post-round metrics:

* Total Punch Count (split by Jabs vs. Power Punches).

* Average Punch Velocity (m/s).

* Guard Hold Percentage (time hands remained near cheeks).

* Evasion Success Rate (percentage of incoming coach attacks slipped/rolled).

* Apple Health / Google Fit sync hook indicator.

5. Curriculum & Level Progression

Level 1: Foundations (Stance & Straight Punches)

* **Focus:** Stance footprints on floor, Guard check, The 1 (Jab) and The 2 (Cross).

* **Combos:** `1-1`, `1-2`, `1-1-2`, `1-2-1`.

* **Pressure Test:** 15 continuous mitt combos with coach without breaking guard.

Level 2: Inside Game & Head Movement

* **Focus:** The 3 (Lead Hook), Slips (Left/Right), Rolls (Under).

* **Combos:** `1-2-3`, `Slip Right - 2`, `1-2-Roll Left-3`, `3 - Step Back`.

* **Pressure Test:** 2-minute round. Coach fires back padded noodle strikes at random intervals; player must duck/slip while maintaining punch output.

Level 3: Body Work & Level Changes

* **Focus:** Level changes (squat mechanics), 3b (Liver shot), 1b/2b (Straight body shots), Parrying.

* **Combos:** `1b-2-3`, `1-2-3b`, `Parry - 2 - 3b - 3`.

* **Pressure Test:** High-Low Mitt Work. Player must squat to hit low targets; standing tall results in counter-strikes.

Level 4: Uppercuts & Angle Creation

* **Focus:** 5 & 6 (Uppercuts), 45-degree Pivots, Pull Counter.

* **Combos:** `1-2-5-2`, `1-2 - Pivot Left - 3 - 2`, `Pull Back - 2 - 3 - 6`.

* **Pressure Test:** Ring Generalship. Coach marches forward; player must pivot and angle out to avoid being backed into virtual ropes.

Level 5: Master Class Flow State

* **Focus:** Feints, Framing/Trapping, No Target UI (pure visual/audio reading).

* **Combos:** `Feint 1 - 2b - 5 - Roll Right - 2`, `1 - Slip Right - 6 - 3 - Pivot Right - 2 - 3b`.

* **The Ultimate Final Exam ("Golden Gloves"):** 3-round continuous sparring simulation without visual target overlays:

* *Round 1:* Speed & Angles (360-degree rapid mitt work).

* *Round 2:* Real-time counter-punching (react to live coach attacks).

* *Round 3:* Freestyle flow (unscripted mitt placement and attacks).

6. Audio Strategy & Copyright-Free Music Architecture

6.1 Music System

* **Copyright Policy:** All background tracks must be 100% Royalty-Free / Creative Commons Zero (CC0) or licensed under non-monetized open usage (e.g., Free Music Archive, YouTube Audio Library royalty-free tracks, or procedural synth soundtracks).

* **Genres:** Energetic Gym Hip-Hop, Phonk, Instrumental Rock, and High-BPM Electronic/Synthwave.

* **Audio Implementation (Web Audio API):**

- * Dynamic Stem System: Music consists of 2 stems (Base Rhythm + Lead Melodic/Drums).
- * During normal drills: Full mix plays at 80% volume.
- * During Coach Speech: Automatic ducking drops music volume by 12dB.
- * During High Exertion / Flow State: High-pass filter engages slightly to emphasize heart rate and breathing audio.

6.2 Spatial Sound Effects (SFX) Requirements

1. **Impact SFX:**

- * `hit\_glove\_heavy.wav`: Hard leather crack for fast cross/hook impact.
- * `hit\_glove\_light.wav`: Soft leather snap for jabs.
- * `hit\_glove\_dull.wav`: Low effort / exhausted punch impact.
- * `mitt\_pop.wav`: Sharp focus-mitt impact sound.

2. **Environment SFX:**

- * `gym\_ambient.wav`: Looped fan hum, distant gym sounds.
- * `bell\_round\_start.wav`: Traditional heavy brass gym bell.
- * `bell\_round\_end.wav`: Triple ring bell.

3. **Voice Lines (Coach):**

- * Audio library categorized by state: `Encouragement`, `Form\_Correction`, `Combo\_Callouts`, `Pressure\_Test\_Warning`.

7. AI / Engineer Implementation Roadmap

When giving this document to an AI coder or engineer, instruct them to build the application in these sequential steps:

1. **Step 1: WebXR & Scene Setup**

- * Initialize a Three.js scene with WebXR VR session support.

- * Add controller rendering and 6DoF hand tracking.
- * Build basic lighting and placeholder gym environment (floor, bag, lights).

2. **Step 2: Physics & Heavy Bag Interaction**

- * Create a heavy bag object suspended with constraint physics.
- * Implement controller velocity measurement on impact.
- * Link punch speed to haptic rumble duration/intensity and spatial impact SFX.

3. **Step 3: Headset Tracking & Defensive Mechanics**

- * Implement headset position tracking to measure slips (X-axis) and rolls/squats (Y-axis).
- * Build visual indicator or coach noodle strike to test evasion mechanics.

4. **Step 4: Coach NPC State Machine & Focus Mitts**

- * Import Coach 3D model with skeletal rigging.
- * Implement state machine for Coach: `Idle`, `Demonstrating`, `Holding\_Mitts`, `Attacking`.
- * Program target detection on virtual focus mitts held by coach.

5. **Step 5: Curriculum Engine & Levels**

- * Build JSON data structures defining combos for Levels 1 through 5.
- * Create sequence checking logic (ensuring punches hit in correct order and tempo).

6. **Step 6: Audio System & Gym Chalkboard Dashboard**

- * Build Web Audio API manager for royalty-free background music and dynamic ducking.
- * Build in-game chalkboard using dynamic canvas textures to render workout stats post-session.